

Umelá inteligencia Zadanie 3

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Cieľ úlohy

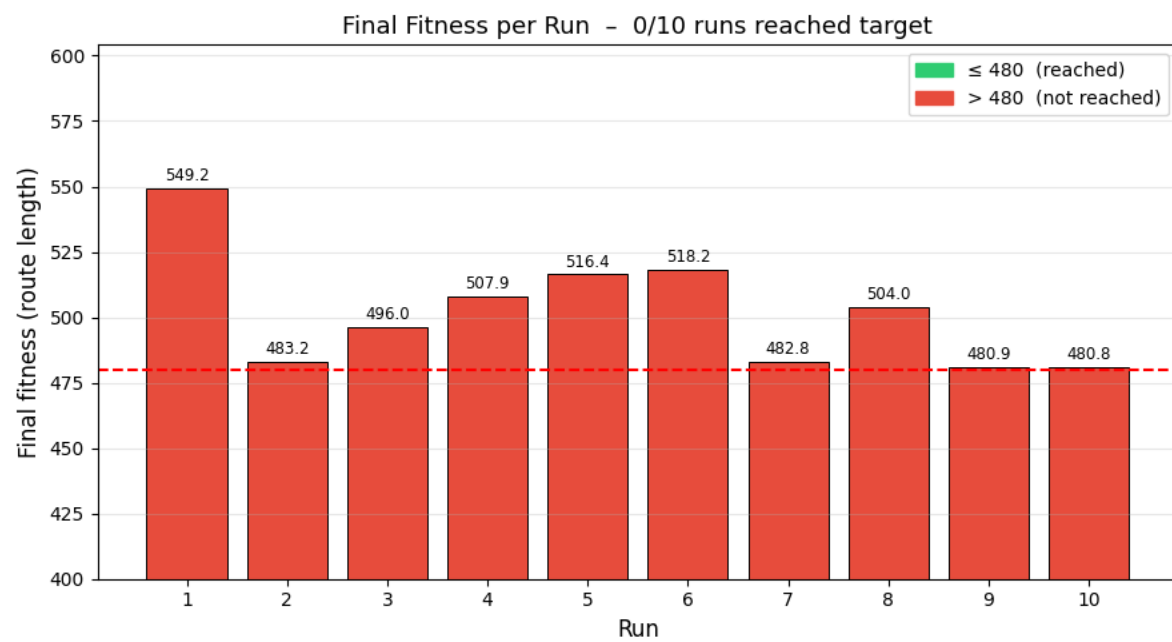
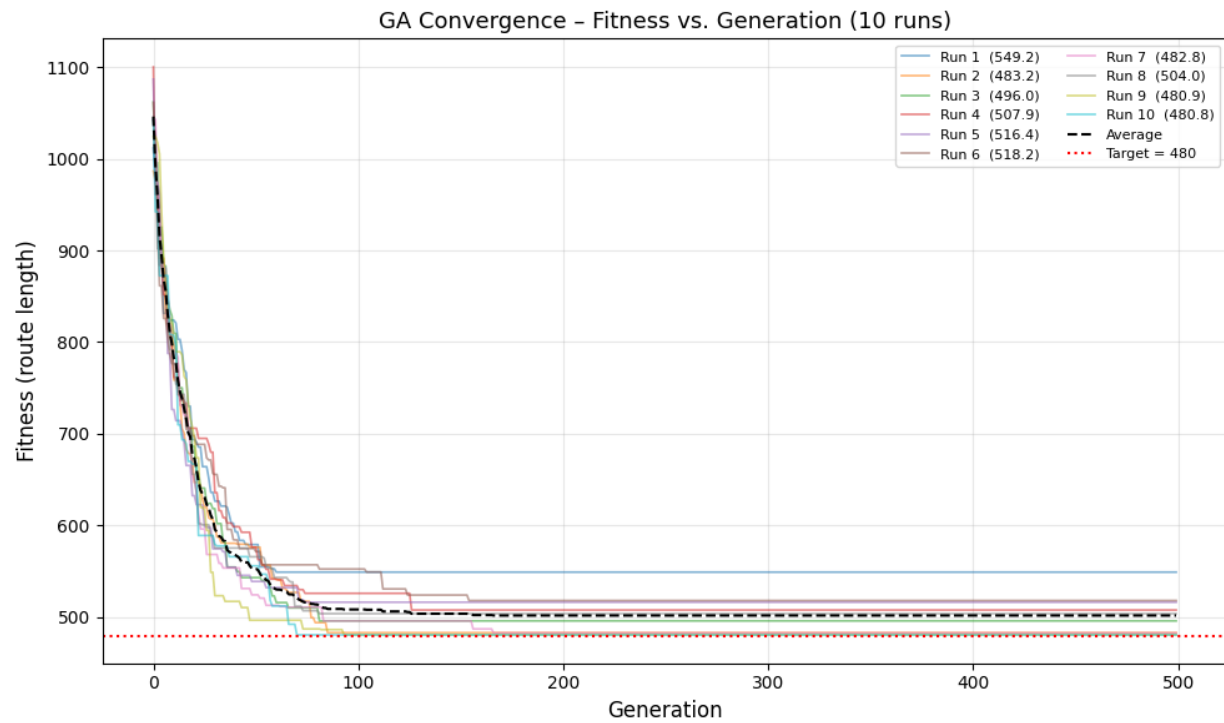
Navrhňte genetický algoritmus pre výpočet najkratšej dráhy mobilného robota, ktorý má prejsť 25 bodov v rovine. Robot začína v bode [0,0] a dráhu ukončí opäť v bode [0,0]. Orientačná kontrolná hodnota: pri vhodnom nastavení GA je možné dosiahnuť dĺžku trasy 468.1095.

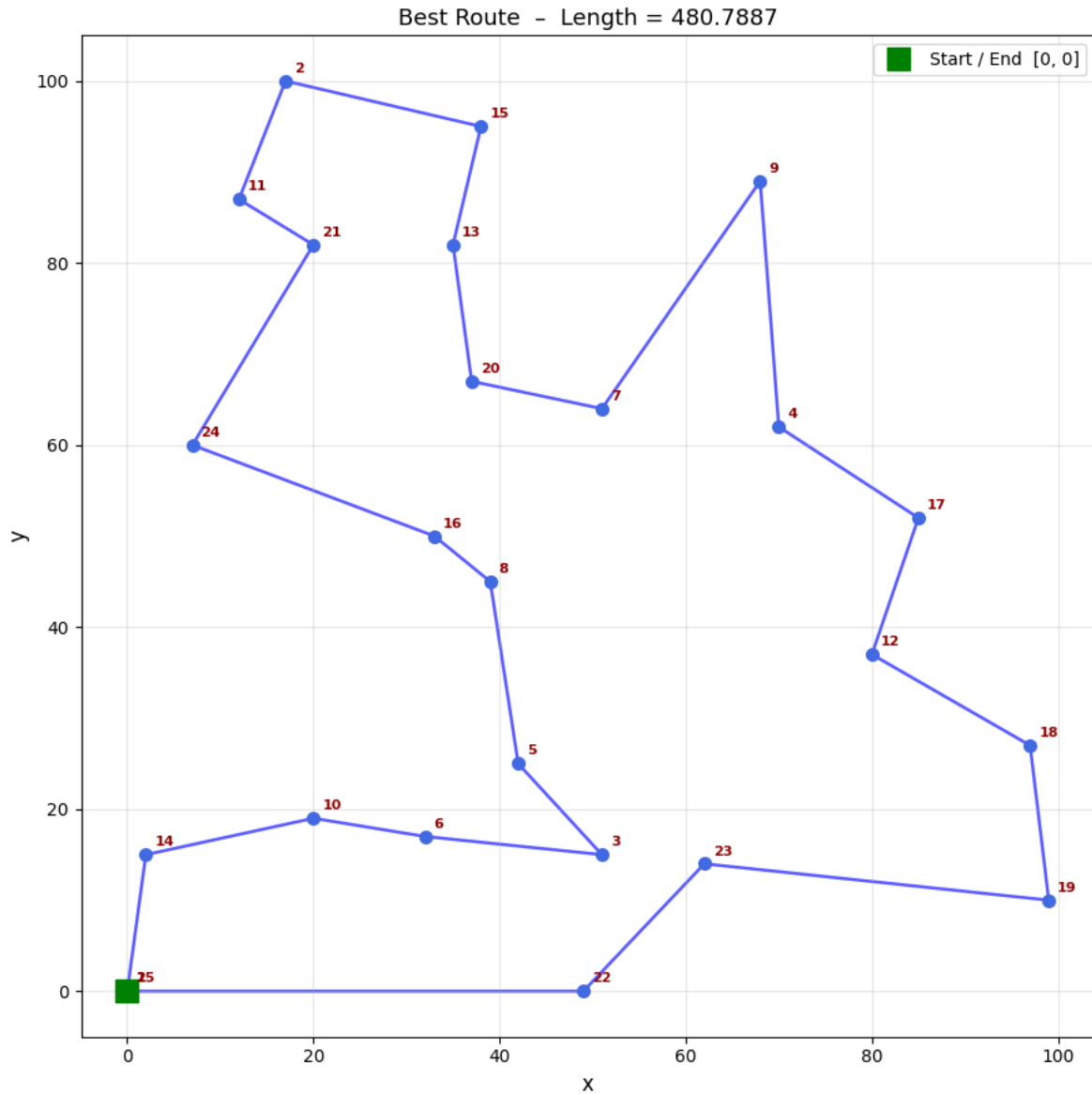
Súradnice bodov [x,y] sú definované v matici B (zoznam dvojíc):

B=

[[0,0], [17,100], [51,15], [70,62], [42,25], [32,17], [51,64], [39,45], [68,89], [20,19], [12,87], [80,37], [35,82], [2,15], [38,95], [33,50], [85,52], [97,27], [99,10], [37,67], [20,82], [49,0], [62,14], [7,60], [0,0]]

```
GA_PARAMS = dict(
    pop_size=200,
    n_generations=500,
    mutation_rate=0.25,
    crossover_rate=0.9,
    elite_count=100
)
```

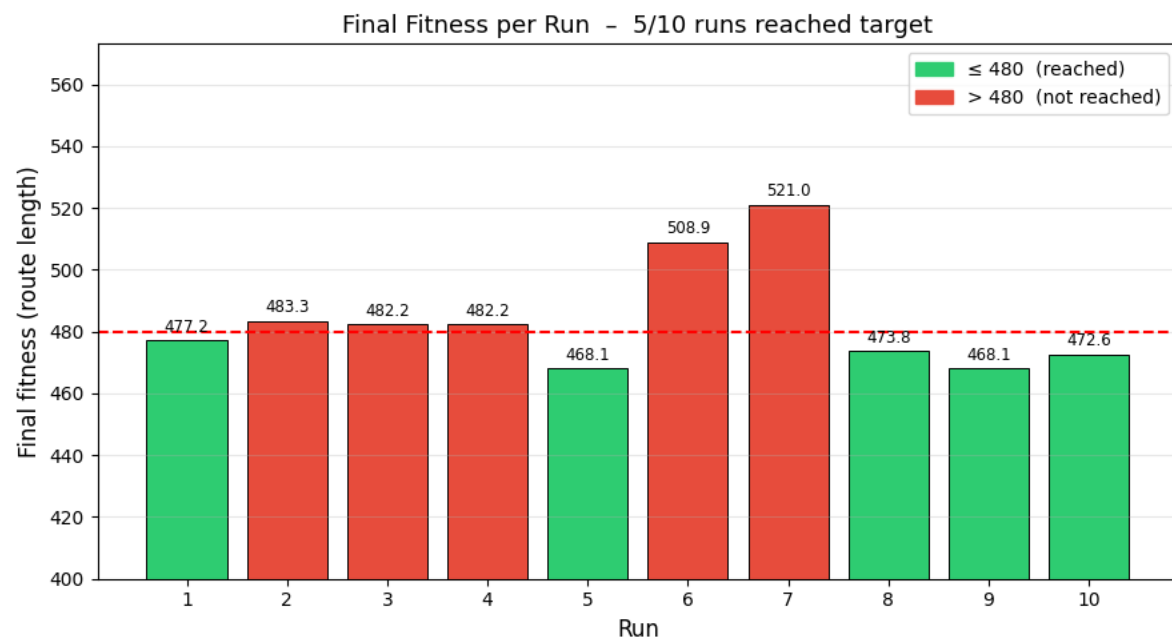
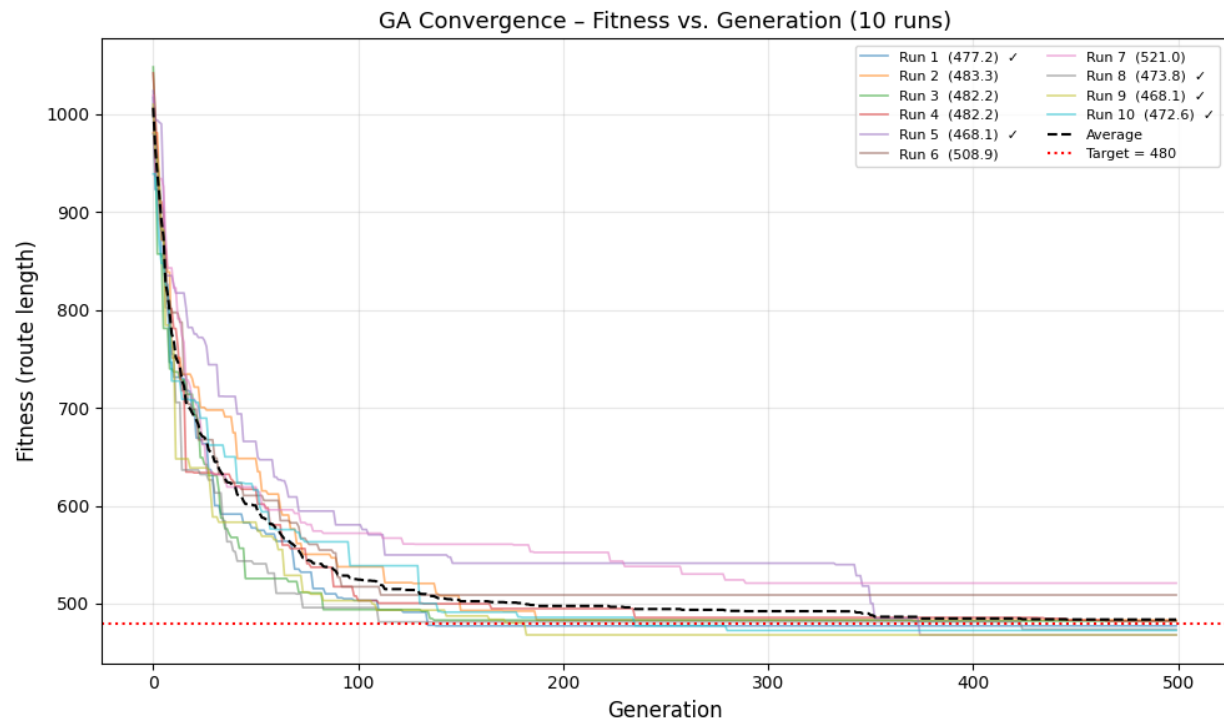


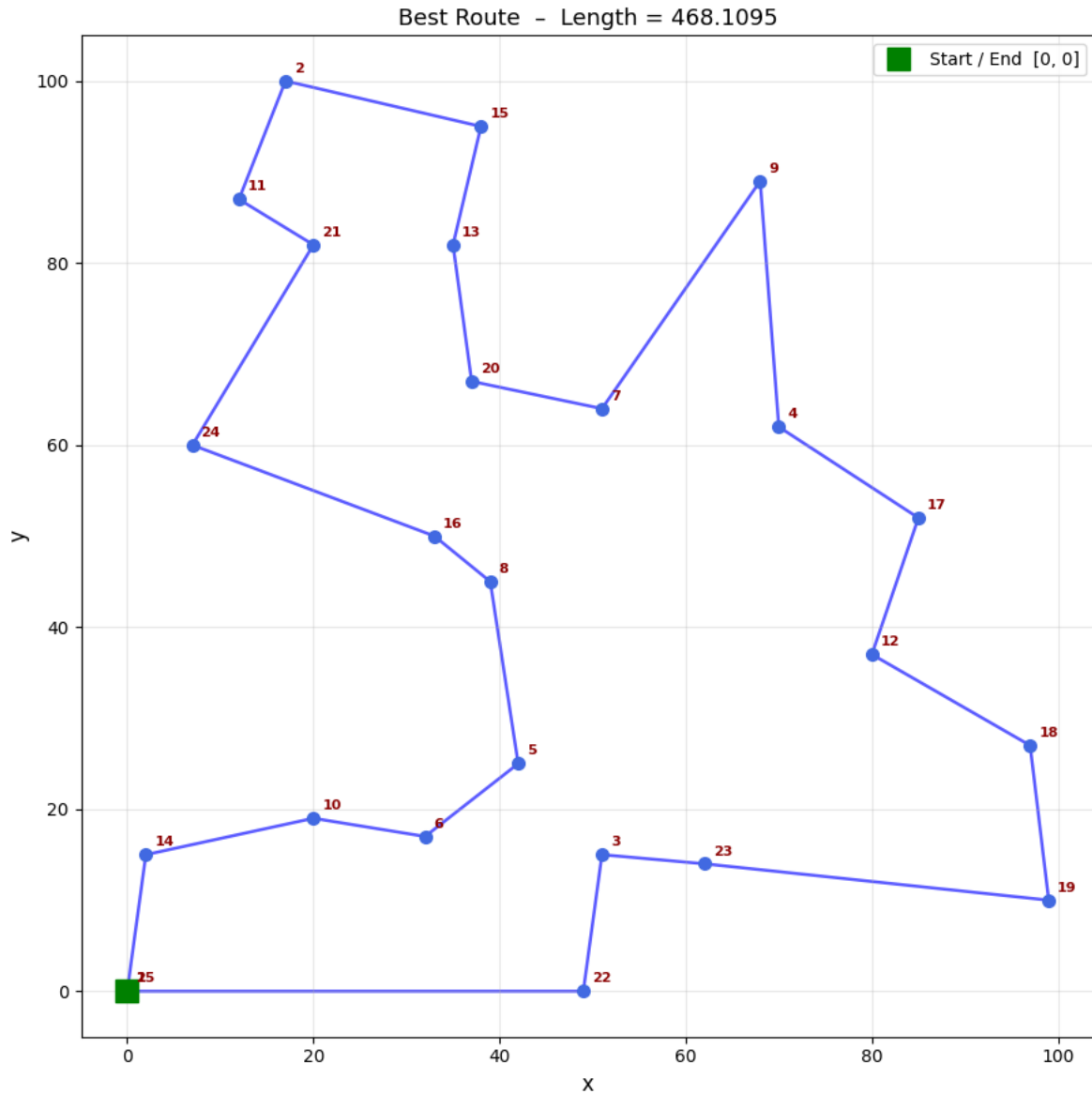


Best genome: [0 21 22 18 17 11 16 3 8 6 19 12 14 1 10 20 23 15 7 4 2 5 9 13 24]

Road: [0, 0] -> [49, 0] -> [62, 14] -> [99, 10] -> [97, 27] -> [80, 37] -> [85, 52] -> [70, 62] -> [68, 89] -> [51, 64] -> [37, 67] -> [35, 82] -> [38, 95] -> [17, 100] -> [12, 87] -> [20, 82] -> [7, 60] -> [33, 50] -> [39, 45] -> [42, 25] -> [51, 15] -> [32, 17] -> [20, 19] -> [2, 15] -> [0, 0]

```
GA_PARAMS = dict(
    pop_size=200,
    n_generations=500,
    mutation_rate=0.60,
    crossover_rate=0.9,
    elite_count=10
)
```

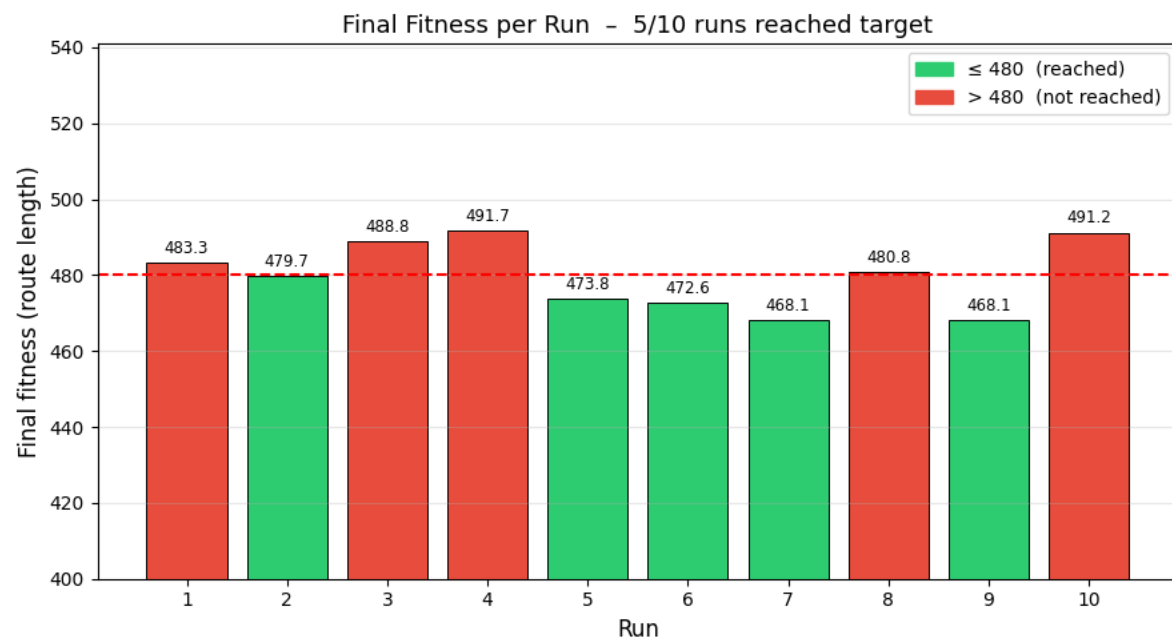
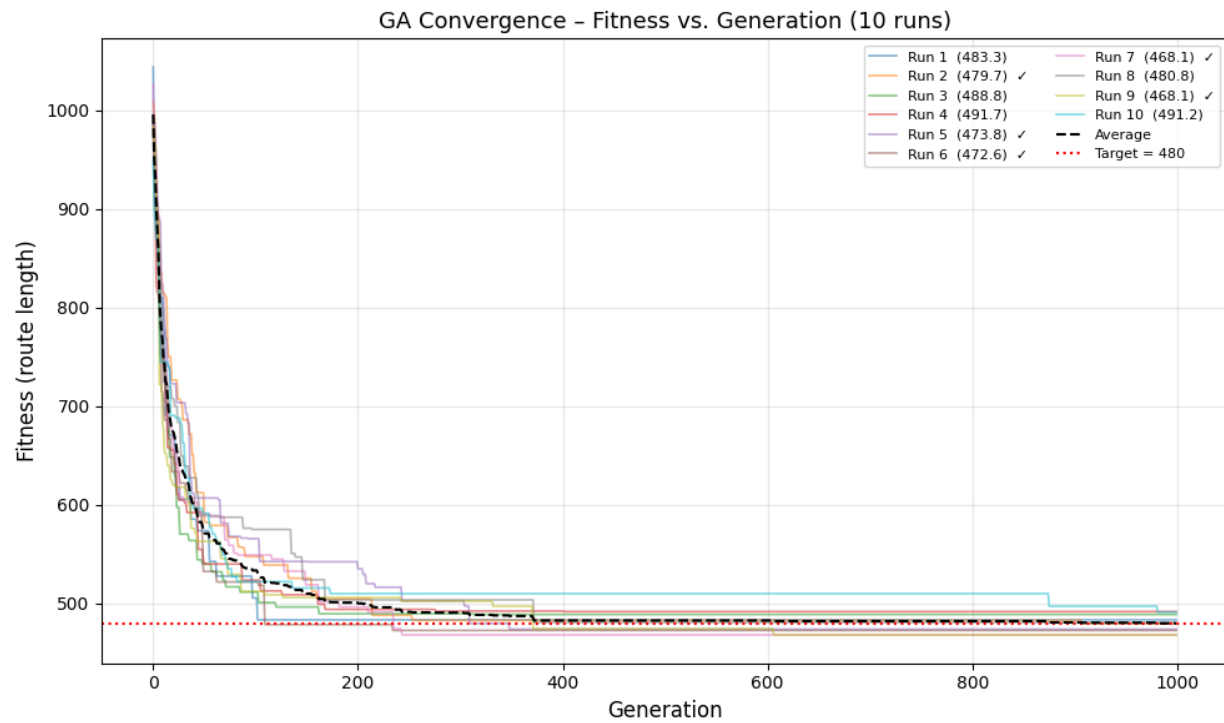


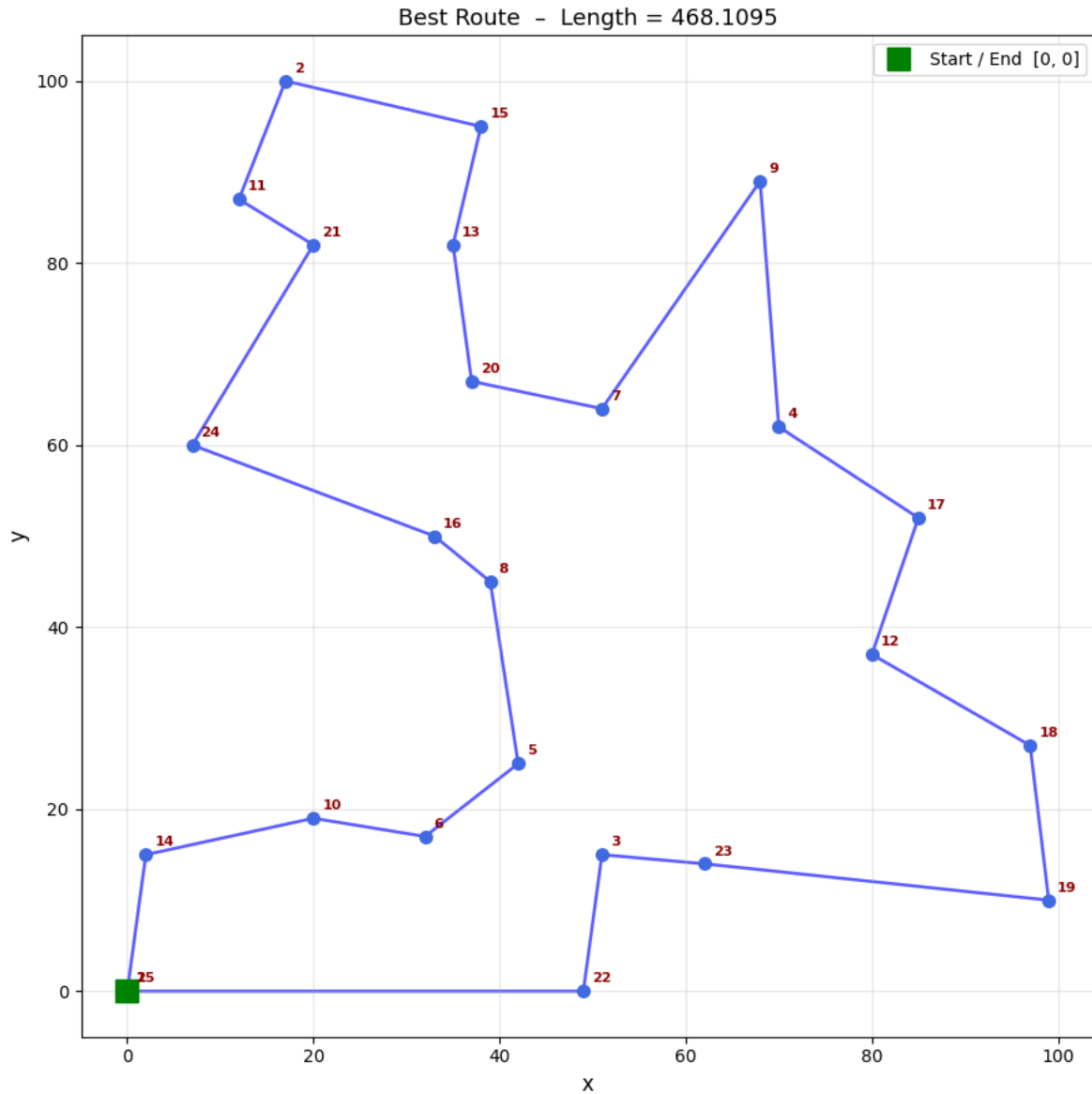


Best genome: [0 21 2 22 18 17 11 16 3 8 6 19 12 14 1 10 20 23 15 7 4 5 9 13 24]

Road: [0, 0] -> [49, 0] -> [51, 15] -> [62, 14] -> [99, 10] -> [97, 27] -> [80, 37] -> [85, 52] -> [70, 62] -> [68, 89] -> [51, 64] -> [37, 67] -> [35, 82] -> [38, 95] -> [17, 100] -> [12, 87] -> [20, 82] -> [7, 60] -> [33, 50] -> [39, 45] -> [42, 25] -> [32, 17] -> [20, 19] -> [2, 15] -> [0, 0]

```
GA_PARAMS = dict(
    pop_size=400,
    n_generations=1000,
    mutation_rate=0.25,
    crossover_rate=0.9,
    elite_count=10
)
```

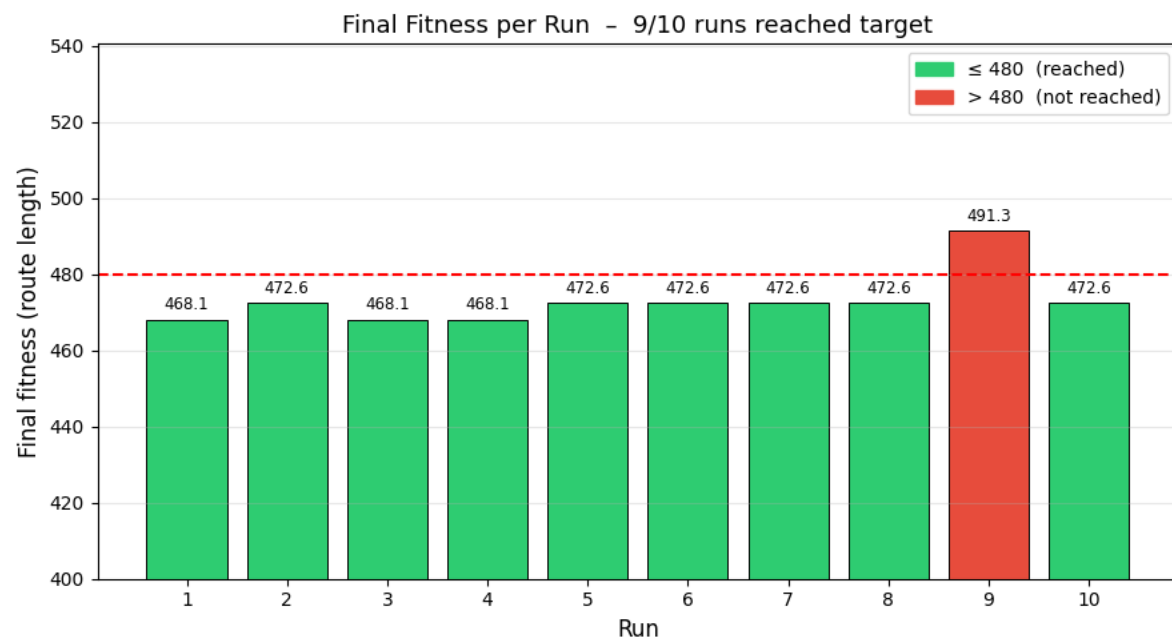
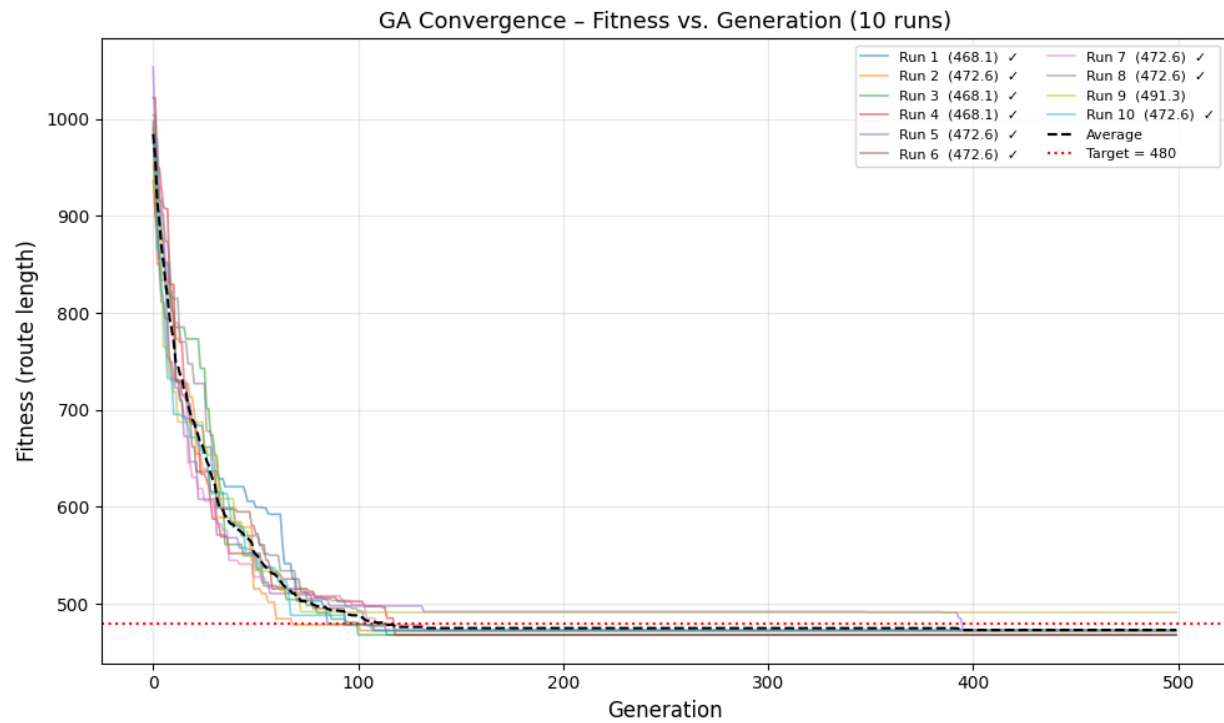


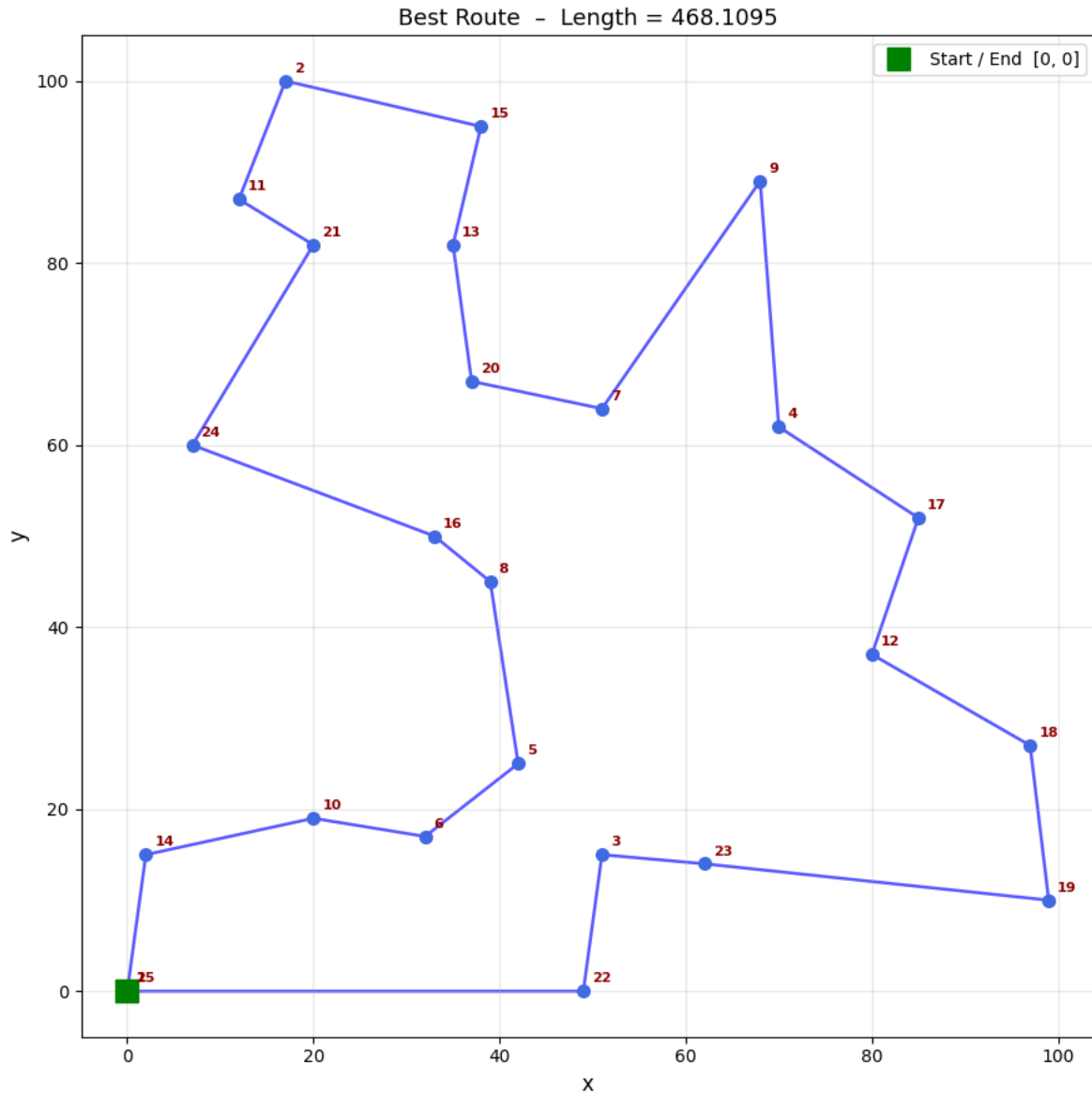


Best genome: [0 21 2 22 18 17 11 16 3 8 6 19 12 14 1 10 20 23 15 7 4 5 9 13 24]

Road: [0, 0] -> [49, 0] -> [51, 15] -> [62, 14] -> [99, 10] -> [97, 27] -> [80, 37] -> [85, 52] -> [70, 62] -> [68, 89] -> [51, 64] -> [37, 67] -> [35, 82] -> [38, 95] -> [17, 100] -> [12, 87] -> [20, 82] -> [7, 60] -> [33, 50] -> [39, 45] -> [42, 25] -> [32, 17] -> [20, 19] -> [2, 15] -> [0, 0]

```
GA_PARAMS = dict(
    pop_size=200,
    n_generations=500,
    mutation_rate=0.25,
    crossover_rate=0.9,
    elite_count=10
)
```





Best genome: [0 21 2 22 18 17 11 16 3 8 6 19 12 14 1 10 20 23 15 7 4 5 9 13 24]

Road: [0, 0] -> [49, 0] -> [51, 15] -> [62, 14] -> [99, 10] -> [97, 27] -> [80, 37] -> [85, 52] -> [70, 62] -> [68, 89] -> [51, 64] -> [37, 67] -> [35, 82] -> [38, 95] -> [17, 100] -> [12, 87] -> [20, 82] -> [7, 60] -> [33, 50] -> [39, 45] -> [42, 25] -> [32, 17] -> [20, 19] -> [2, 15] -> [0, 0]